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Microaeration enhances methane production in anaerobic digestion: A systematic review and meta-analysis

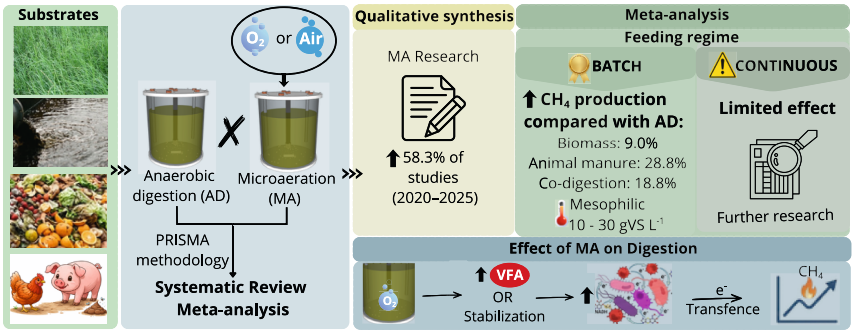
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HIGHLIGHTS

- Research on microaeration (MA) in anaerobic digestion has increased since 2010.
- Microaeration significantly increased methane production in batch systems.
- Biomass, manure, and co-digestion showed the greatest methane increases under MA.
- Methane enhancement depended on oxygen input and operational conditions.
- Meta-analysis identified key parameters for MA optimisation and scale-up.

GRAPHICAL ABSTRACT



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ABSTRACT

Microaeration (MA) has emerged as a promising strategy to intensify anaerobic digestion (AD) of solid waste and wastewater. This systematic review and meta-analysis aimed to evaluate the operational parameters most frequently applied in MA and their effects on methane production in comparison to conventional AD. The Scopus, Web of Science, and PubMed databases were searched, yielding 47 studies eligible for a meta-analysis. The qualitative analysis revealed that research on this strategy has increased in recent years, with 50.0 % of the studies published between 2020 and 2025. However, most studies have been conducted at a small scale, with a predominance of batch experiments and direct application in the liquid phase, which limits extrapolation of the results to real operational conditions. Based on the meta-analysis, MA promotes a significant overall increase in methane production in batch systems, with an average gain of 9.0 % for biomass, 28.8 % for animal manure, and 18.8 % for co-digestion, particularly under mesophilic conditions and intermediate volatile solids concentrations (10–30 g L^{-1}). In continuous-feeding reactors, MA has a relatively limited, non-significant effect; nevertheless, it markedly increases biomass (+6.6 %). Overall, the results indicate that MA has clear potential to intensify rate-limiting steps of AD; however, its effectiveness is highly dependent on the interaction between substrate characteristics and operational conditions. Its broader implementation therefore requires greater standardisation of oxygen input and control strategies, improved data reporting, and validation at pilot and large scales under realistic treatment conditions.

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1. Introduction

Anaerobic digestion (AD) is a well-established technology for wastewater and solid waste treatment, bioenergy generation, and resource recovery from diverse substrates (Duan et al., 2025). Beyond methane production digestate use as biofertiliser, AD pathways can be redirected towards high-value products, such as volatile fatty acids (VFAs) and biohydrogen, by suppressing methanogenic activity (Feng et al., 2022). This versatility supports the transition of AD from a waste-treatment technology into a resource-recovery platform aligned with a circular economy (United Nations [UN], 2015).

Despite this potential, AD is a complex biochemical process governed by precise microbial syntrophy and remains sensitive to operational disturbances. Key limitations include low hydrolysis rates – commonly the rate-limiting step – and hydrogen sulfide (H_2S) formation, which leads to corrosion of equipment (Chen et al., 2020; Ullah et al., 2025). In addition, process instability may arise from variations in temperature, the organic loading rate, and pH. Therefore, to improving biogas production, it is necessary to develop strategies that enhance hydrolysis, as they can reduce the hydraulic retention time while increasing energy recovery (Huilinir et al., 2023).

Microaeration (MA), defined as the controlled injection of small amounts of air or oxygen (0.005–5 L O_2 per L reactor per day), has emerged as a promising strategy to mitigate the inherent limitations of AD (Nguyen and Khanal, 2018). Compared with conventional AD, MA promotes microbial diversity and syntrophic interactions (Chen et al., 2020). The reported benefits include an H_2S removal efficiency greater than 85 % (Díaz et al., 2011; Kraakman et al., 2023) and a marked increase in hydrolysis (up to 233 %) (Chetawan et al., 2021). Moreover, MA increases biogas and methane production by up to 27 % and 211 %, respectively (Lim and Wang, 2013; Montalvo et al., 2016), while improving the overall methane yield by approximately 24 % (Di et al., 2025). These effects are associated with stimulation of hydrolytic and acidogenic microorganisms, such as Aminicenantes (Chuenchart et al., 2025), and the metabolic reinforcement of hydrogenotrophic methanogenesis (Zhu et al., 2022).

In the literature, MA is described using different terms, including oxygenation, limited aeration, microaerophilic conditions, and micro-oxygenation (Bao et al., 2020; Chen et al., 2020; Oliveira et al., 2021a). The oxygen input, the main operational parameter, is reported using different metrics depending on substrate and reactor configuration, such as volumetric loading rates (L O_2 per L reactor per day) (Nguyen and Khanal, 2018), normalisation to volatile solids (L O_2 per g volatile solids [VS]) (Song et al., 2021), and real-time indicators including dissolved oxygen (0.1–1 mg L^{-1}) and the oxidation–reduction potential (ORP; –300 to 0 mV) (Ullah et al., 2025). Aeration intensity is also expressed as volume of air per volume of liquid per minute (vvm) (Huilinir et al., 2023).

Although several reviews have summarised MA applications in AD (Chen et al., 2020; Huilinir et al., 2023; Nguyen and Khanal, 2018; Ullah et al., 2025), no study has quantitatively integrated these findings or assessing their effects on methane production through a meta-analysis. A meta-analysis synthesises disparate data into an integrated quantitative evaluation. This approach would enable a robust assessment of how MA impacts energy recovery, aligning with the mechanistic improvements in methane production reported by Xu et al. (2024).

To address this gap, this systematic review and meta-analysis aimed to quantify the impact of MA operational parameters on methane production compared with conventional AD. Specifically, it (i) identified key parameters, including substrate type, experimental scale, the VS concentration, oxidising agents, the MA rate, frequency, and oxygen application location; and (ii) quantified their effect sizes on methane production in batch and continuous systems. The results provide a comparative framework to support the application of MA in sustainable waste treatment.

2. Materials and methods

This systematic review followed the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) (Page et al., 2021). The process comprised study identification, screening, eligibility assessment, and inclusion. These steps were applied to both the qualitative synthesis and the quantitative assessment, the latter conducted as a meta-analysis.

2.1. Study identification and selection

The Scopus (Elsevier), Web of Science (Clarivate Analytics), and PubMed (National Library of Medicine) databases were searched for studies related to the use of MA in AD published up to June 2025. The search filters were set to 'all fields' for Web of Science and PubMed, and to 'title, abstract, or keywords' for Scopus. The searches were performed in English using the following search string: ('anaerobic digestion') AND ('microaerat*' OR 'microaerophilic' OR 'micro-aerat*' OR 'limited aerat*' OR 'micro-oxygen*' OR 'oxygenation' OR 'microaerobic' OR 'limited oxygen*' OR 'moderate oxygen*' OR 'oxygen-limit*' OR 'micro-oxygen*' OR 'air injection' OR 'oxygen* supply'). Duplicate studies were automatically removed using R version 4.3.3 (R Core Team, 2024).

The selection process comprised two stages. In the first stage, the titles and abstracts of all retrieved articles were screened to identify those with potential to meet the eligibility criteria. This screening step employed machine learning through interactive text classification using the open-source tool ASReview (Van de Schoot et al., 2021). To train the model, 30 relevant and 30 irrelevant papers were manually selected, and the Naïve Bayes algorithm was applied for ranking (Bernardes et al., 2022). Once trained, the model ranked all papers in decreasing order of relevance. After identifying 50 consecutive irrelevant articles, the remaining articles were assumed to be irrelevant, concluding the first stage. In the second stage, the full texts of all pre-selected papers were assessed to confirm that they met the eligibility criteria.

2.2. Eligibility and inclusion

The eligibility criteria were defined according to the PICO framework: P – population, I – intervention, C – comparator/gold standard, and O – outcome (Page et al., 2021). The population of interest comprised original research articles published in English that focused on AD and explicitly specified the substrates used. The intervention was MA applied to AD. The comparator was conventional AD. The outcome of interest was methane production.

Furthermore, studies were included only if they applied controlled MA, enabling quantification of the supplied air or oxygen. In addition, anaerobic and microaerobic conditions had to be maintained under identical operational settings to isolate the effect of MA from other factors influencing methane production. To be included in the meta-analysis, the study had to report basic experimental data, including the sample size and mean methane production. When the standard deviation was not reported, it was assumed to be 10 % of the mean (Xu et al., 2024).

The PRISMA flow diagram in Fig. 1 shows the number of studies selected at each stage, including those retained for the qualitative synthesis and the meta-analysis. The database searches yielded 961 records. After applying database filters and removing duplicates, 581 records proceeded to screening. Following title and abstract screening, 148 records were selected for full-text assessment. Ultimately, 58 studies met the eligibility criteria and were included in the qualitative synthesis (Table S1), corresponding to 271 independent cases. For the meta-analysis, 47 studies were included, representing 228 independent cases.

2.3. Data extraction

The following data – relevant for AD performance and operational

control – were extracted data from the selected studies: (i) the substrate type; (ii) the VS concentration; (iii) the operating temperature; (iv) the study scale; (v) the MA rate; (vi) the MA operational conditions, including the oxygen input and source, strategy, site, and frequency; (vii) mean methane production; (viii) the sample size; and (ix) the standard deviation. Data reported exclusively in graphical form were digitised using the online application WebPlotDigitizer, version 4.8 (Rohatgi, 2023).

For both the qualitative synthesis and the meta-analysis, data were organised into groups based on the substrate type (Table S2). Oxygen inputs were normalised to the VS concentration (Table S3), as this was the most used control parameter among the selected studies. When oxygen inputs were reported as air supply, an oxygen content of 20 % was assumed to convert values to oxygen equivalents (Oliveira et al., 2021a).

The following categories were adopted for the MA application method: (a) pretreatment, in which MA was applied in a separate unit before conventional AD or added to the reactor up to the fifth experimental day (batch experiments); (b) during digestion, where MA was maintained throughout the entire digestion period within the reactor; and (c) combined strategy, with the application of MA both as a pretreatment and during digestion. The frequency of MA was classified as: (a) single, when MA was applied only once; (b) intermittent, when MA was applied periodically; and (c) continuous, when air or oxygen was continuously injected during the microaerobic phase.

2.4. Statistical analysis

A meta-analysis was conducted to evaluate the effect of MA on methane production across different parameters, compared with conventional AD. Data were extracted from controlled bioassays, in which a single study could include multiple MA conditions tested in independent experiments, each treated as an independent case.

The effect size for methane production was calculated using the natural logarithm of the response ratio (ln RR) between the treatment

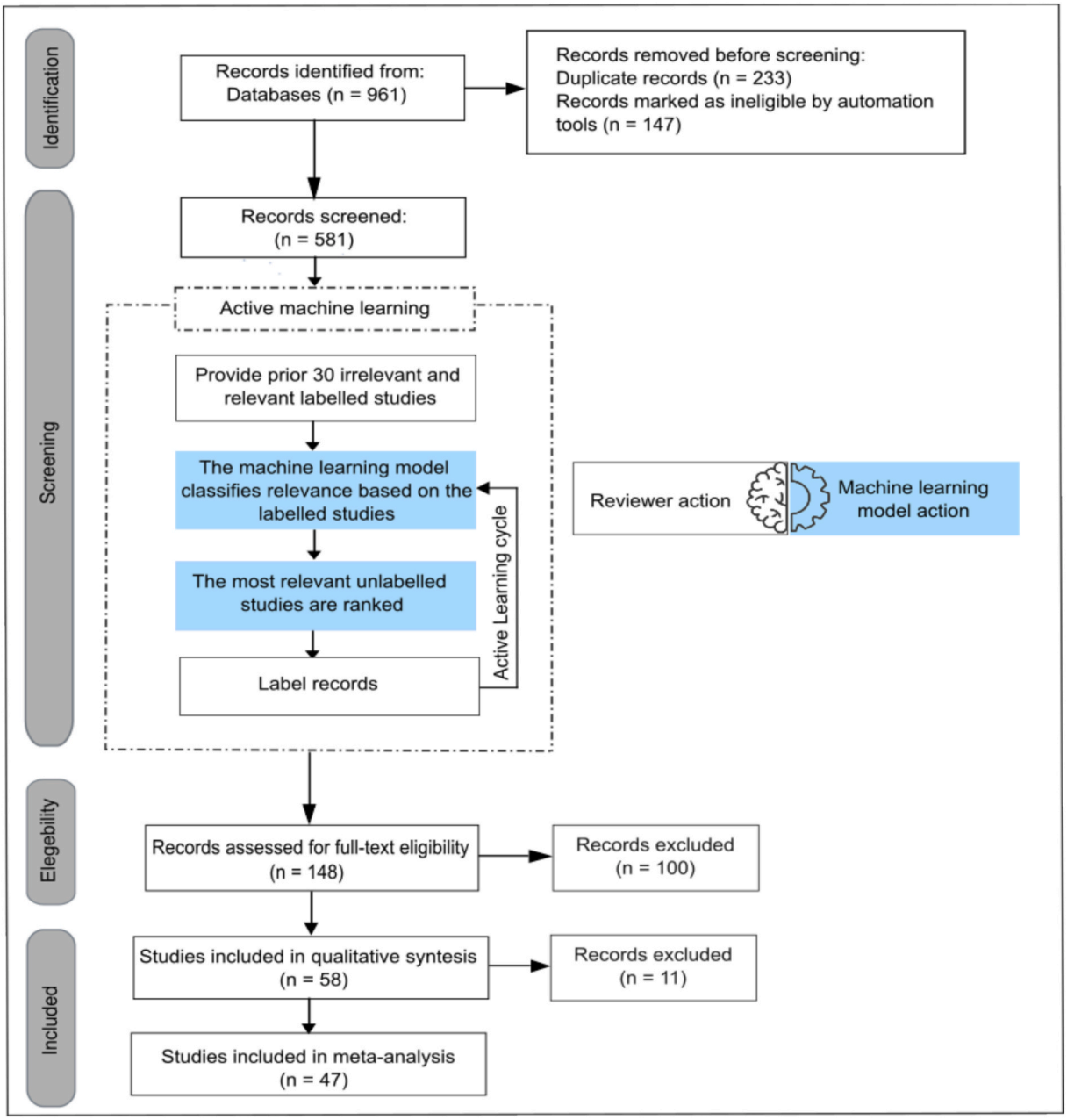


Fig. 1. PRISMA data collection flowchart.

mean, MA (X_M), and the control mean, conventional AD (X_C). The calculation was performed following Hedges et al. (1999) (Equation (1)), using a mixed-effects model.

$$lnRR = ln(X_M/X_C) \quad (1)$$

Statistical analyses were conducted using R version 4.3.3 (R Core Team, 2024) with the meta package. The estimated effects are presented with 95 % confidence intervals (CIs). Between-study heterogeneity (τ^2) was estimated using the method described by Dersimonian and Laird (1986). Total heterogeneity was assessed using the I^2 statistic, considered significant at $p \leq 0.05$ (Jackson et al., 2012). Heterogeneity was classified as low when I^2 was ≤ 25 % and high when I^2 was ≥ 75 %.

The results are presented as forest plots. For interpretation, values less than 0 (left side of the plot) indicate a negative effect of the intervention, meaning methane production under MA was lower than under conventional AD. Conversely, values greater than 0 (right side) indicate that MA had a positive effect, with higher methane production compared with the anaerobic control. Additionally, effect sizes are considered statistically significant ($p \leq 0.05$) when the 95 % CI of the lnRR does not include zero.

3. Results and discussion

3.1. Qualitative analysis

3.1.1. Overview of the included studies

As shown in Fig. 2, among the studies included for the qualitative analysis, there was a progressive increase in scientific output related to the effect of MA on methane production, with more continuous production beginning in 2010. Fifty per cent of the studies were published between 2020 and June 2025, indicating growing interest in this strategy within the context of AD. This trend suggests advancement in research on the use of MA for waste treatment, reflecting the need to identify approaches that enhance energy conversion through the digestion of different substrates.

MA can be applied as a pretreatment, throughout digestion, or in a combined approach – pretreatment and then during digestion (Zhan et al., 2023). When used as a pretreatment, the aim is to promote facultative microorganisms and to increase the biodegradability of organic matter (Ullah et al., 2025). During digestion, MA supports process stability, primarily through the control of VFAs (Chen et al., 2020). The use of both strategies has increased over time (Fig. 2). Although pretreatment is more recent (with the first study published in 2013), it accounts for 52.8 % of the total cases, while application during digestion represents 46.5 % of the total cases. The growing interest in pretreatment may be related to the need for more precise control of oxygen input, as excess oxygen can reduce methane potential, whereas insufficient oxygen has no effect (Girotto et al., 2018).

Fig. 3 presents the distribution of cases across the substrate categories, experimental scale, MA application strategy, oxidising agent, and the basis for MA quantification. MA has been applied to a range of urban and agricultural substrates (Ullah et al., 2025).

Among the evaluated groups, biomass – primarily composed of lignocellulosic agricultural residues – was the most studied, accounting for 40.2 % of the cases (Fig. 3A). The extensive use of these residues is related to their low biodegradability, which results from the rigid structure of cellulose, hemicellulose, and lignin (Xu et al., 2024; Yu et al., 2019). There were fewer cases for the other substrate groups, particularly for co-digestion studies, highlighting opportunities for future research.

Most experiments were conducted at small scales, with the predominance of the bench (69.4 %) and demonstration (18.1 %) scales, with fewer cases at the pilot (10.3 %) and large (2.2 %) scales (Fig. 3B). Large-scale research has mainly focused on H_2S removal from biogas (Ullah et al., 2025), which explains the limited number of studies directly targeting methane production. This scenario indicates that MA remains under development as a technology for enhancing energy yield.

There has been considerable variability in the technical parameters of MA in anaerobic reactors, including when it is applied, the oxygen source, and the basis for MA quantification. Regarding the application strategies, 39.3 % of the cases used a single application, 34.4 % used intermittent application, and 26.3 % used continuous application (Fig. 3C). It is necessary to compare different application strategies to establish more robust operational conditions (Chen et al., 2020; Ullah et al., 2025). The choice of oxygen source – pure O_2 or air – depends on both the digestion objective and operational costs (Chen et al., 2020). Despite its higher cost, pure O_2 was the most commonly used source, applied in 55.7 % of the cases (Fig. 3D). Air injection (approximately 80 % N_2 and 20 % O_2) can dilute biogas, especially when applied to the reactor headspace (Nguyen and Khanal, 2018). Therefore, comparative evaluation of oxygen sources is important to assess MA performance and

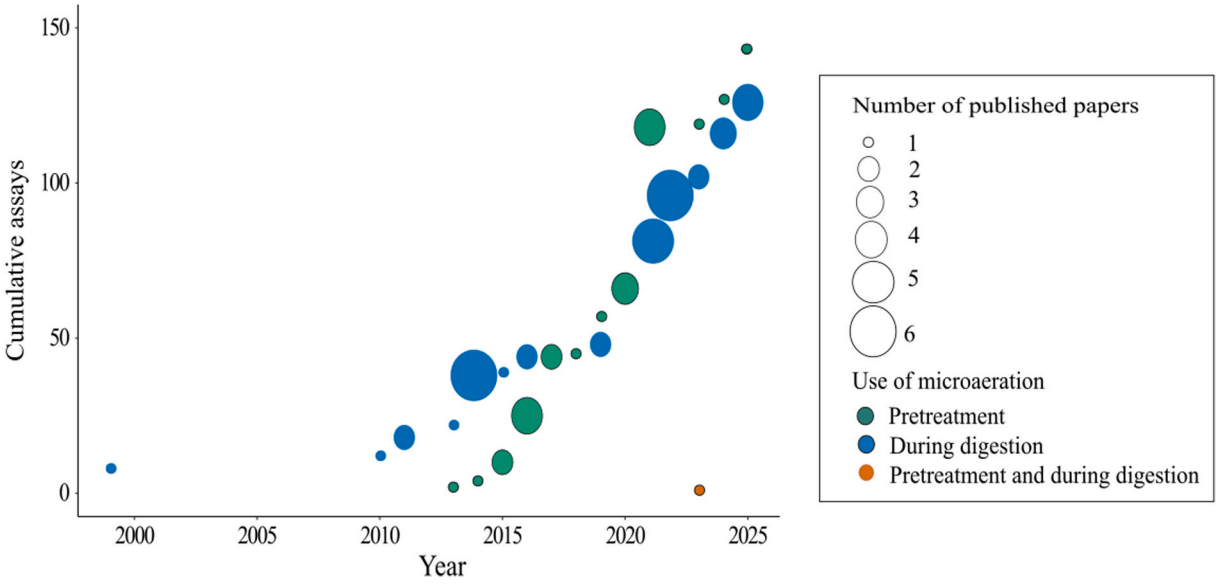


Fig. 2. The number of studies included in the review and the number using microaeration as a pretreatment, during digestion, or as a pretreatment and during digestion.

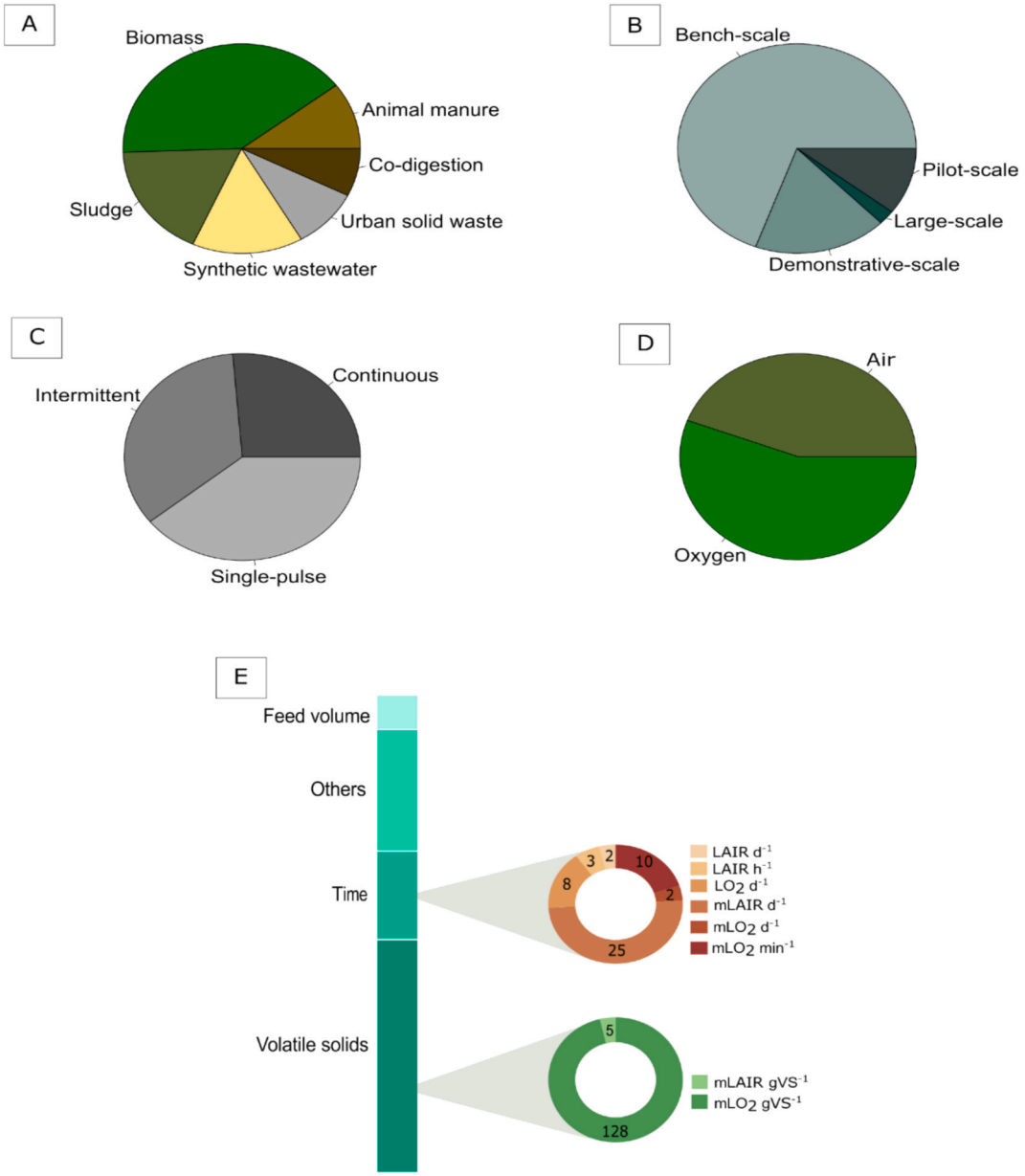


Fig. 3. The (A) substrates, (B) study scale, (C) microaeration application, (D) microaeration agent, and (E) the basis for microaeration quantification.

to support the development of control models and more efficient strategies (Ullah et al., 2025).

The rate at which air or pure O_2 is applied is a key factor for MA performance: it directly influences efficiency across different substrates and reactor types (Nguyen and Khanal, 2018). Due to its importance, various parameters have been used to quantify MA, including the VS concentration, feed volume, application time, and other specific methods (Fig. 3E). Among these, the VS concentration was the most frequently used parameter (49.1 % of the cases). Other approaches were used for 25.5 % of the cases; these include units based on substrate or reactor volume, pressure, biogas production, vvm, the ORP, volume of air per gram of total solids per minute (vmm), as well as daily MA application not linked to a specific operational parameter.

3.1.2. *Interactions among the parameters in the included studies*

It is also relevant to evaluate the relationship between the physicochemical characteristics of the substrates and the MA operational parameters (Figs. S1, S2 and 4). Among the included cases, MA was almost exclusively investigated as a pretreatment at the bench scale (97.9 %). On the other hand, the application of MA during digestion was investigated at the bench scale (37.3 %), demonstration scale (35.7 %), pilot scale (19.8 %), and large scale (4.8 %) (Fig. S1). This distribution suggests a more advanced stage of development for the application of MA during digestion; however, both approaches still require further validation at larger scales to allow direct comparisons (Ullah et al., 2025). At the bench scale, dose control, particularly in biochemical methane potential (BMP) assays, is most frequently based on VS concentration (66.8 % of the cases) because these cases usually involve detailed substrate characterisation. The VS concentration has also been used in some demonstration-scale cases (16.7 %). However, the applicability of the VS concentration decreases as the scale increases due to operational variability and limited monitoring capacity (Chen et al., 2020; Ullah et al., 2025). As a result, larger-scale studies more frequently adopt time-and volume-based strategies to quantify MA (Fig. S1).

Most cases applied MA as a pretreatment and during digestion directly to the liquid phase (80.2 %), consistent with the focus on the methane yield, whereas headspace application (19.8 %) is mainly associated with H_2S removal (Fig. S2). For the liquid phase, a single

application of MA was the most frequently adopted strategy (42.9 % of the cases), likely reflecting the high number of batch pretreatment cases, followed by intermittent application (40.5 %). Intermittent MA, typically one application of MA each day, may improve control of the oxygen supply and VFA accumulation. In contrast, headspace application is typically continuous (59.0 %), as it was in 92.3 % of the cases using recirculation systems, mainly targeting H_2S removal and VFA control (Fig. S2). Moreover, comparative evidence suggests that intermittent MA may enhance the methane yield relative to both conventional AD and single-application strategies (Zhan et al., 2023).

Among the substrate subgroups, VS1 ($\leq$10 g L^{-1}) was the most frequent, followed by VS2 (>10–30 g L^{-1}) and VS3 (>30–100 g L^{-1}), representing 39.1 %, 29.1 % and 28.2 % of the cases, respectively (Fig. 4). There was a similar trend regarding the oxygen input, with lower intensities predominating: the G1 subgroup ($\leq$ 10 mL O_2 g VS^{-1}) accounted for 67.0 % of the cases, while the G2 subgroup (>10–30 mL O_2 g VS^{-1}) represented 17.7 %. The prevalence of low oxygen input remained consistent across the VS concentration subgroups, with the G1 subgroup predominating in all of them: 64.2 % for VS1, 72.9 % for VS2, 59.2 % for VS3, and 100.0 % for VS4 (>100 g L^{-1}). For the G1, G2, and G3 subgroups, a single application of MA was the most frequently applied strategy, accounting for 34.0 %, 76.3 %, and 66.7 % of the cases, respectively. Taken together, selection of the oxygen input and the MA application strategy is directly linked to the objective of the specific study: to enhance hydrolysis, to mitigate the formation of toxic compounds, or to prevent oxygen accumulation (Rashvanlou et al., 2020).

Overall, the qualitative analysis indicates that MA has progressed across different substrates, scales, and application strategies, although most experiments are conducted at the bench scale with a low oxygen input directly into the liquid phase. This predominance of small-scale experiments limits a comprehensive understanding of the potential advantages of MA for methane production under realistic operational conditions, particularly regarding process stability, energy balance, and long-term performance. Similar trends have been reported in previous assessments of emerging AD strategies, where scale-related constraints have hindered the extrapolation of results to pilot and large-scale systems (Chen et al., 2020; Huilinir et al., 2023). The marked heterogeneity in the operational parameters across the included studies also reflects the exploratory nature of MA development and reinforces the need for greater standardisation and systematic evaluation across scales.

These limitations can be overcome by applying a meta-analysis to simultaneously assess of the effects of substrate characteristics, MA application strategy, oxygen input, the oxidising agent, and the operational scale on methane production. The resulting quantitative evidence cannot be obtained from individual studies (Xu et al., 2024). By clarifying the relative importance and interactive effects of these variables, such analyses can inform the development of more robust experimental designs and support the optimisation of operational conditions for subsequent pilot- and full-scale investigations.

3.2. Meta-analysis

A quantitative meta-analytical approach was employed to evaluate the effect of digestion parameters and MA conditions on methane production performance in microaerobic systems compared with conventional AD. To account for the heterogeneity among the included studies and the fact that the feeding strategy directly influences methane production (Rosas-Mendoza et al., 2021), the meta-analysis was divided into two groups: batch studies and continuous feeding studies. This classification enabled a more consistent assessment of MA effects under distinct operational configurations.

3.2.1. The effect of MA on methane production in batch systems

The forest plots in Fig. 5 summarise the effect of MA on methane production in batch studies, distinguishing between substrate-related parameters (Fig. 5A) and operational conditions (Fig. 5B). Overall, MA showed a positive and significant effect on methane production across the substrate and temperature groups, with an identical lnRR (0.08) and 95 % CI (0.05–0.11). There was a similarly positive effect for the VS concentration (lnRR = 0.07, 95 % CI 0.04–0.10). Heterogeneity among studies was high (I^2 > 75 %), indicating substantial methodological and experimental variability.

When disaggregated by substrate type, there were significant increases in methane production for biomass, animal manure, and co-

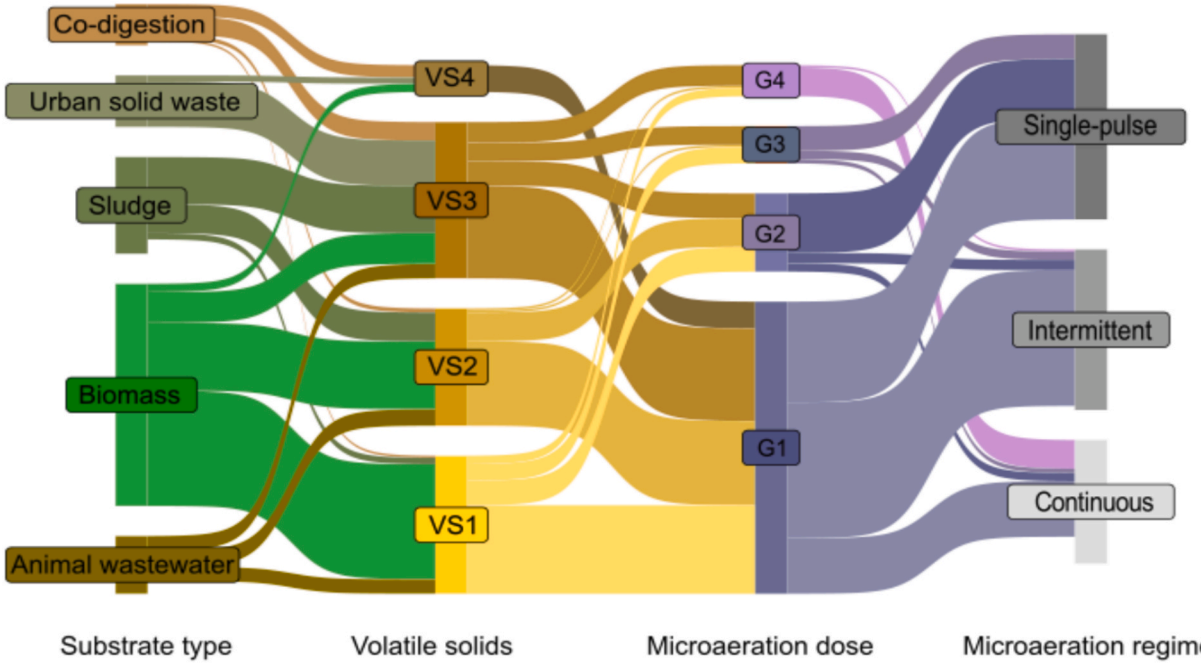
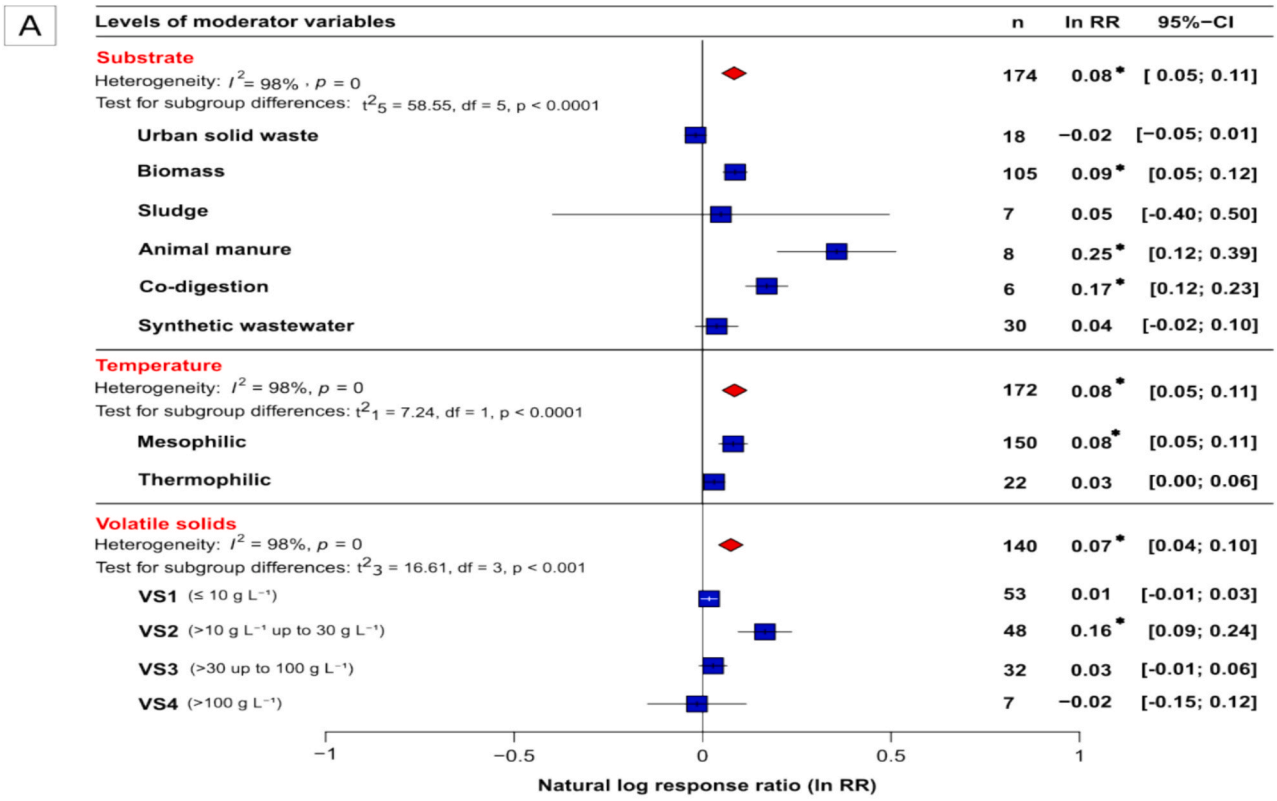


Fig. 4. Interaction flows among the substrate type, volatile solid concentration, oxygen input (i.e., microaeration dose) and application strategy (i.e., regime). Abbreviations: VS1, volatile solids $\leq$ 10 g L^{-1}; VS2, volatile solids > 10 up to 30 g L^{-1}; VS3, volatile solids > 30 up to 100 g L^{-1}; VS4, volatile solids > 100 g L^{-1}; G1, $\leq$ 10 mL O_2 g VS^{-1}; G2, > 10 up to 30 mL O_2 g VS^{-1}; G3, $\geq$ 30 up to 60 mL O_2 g VS^{-1}; G4, > 60 mL O_2 g VS^{-1}.



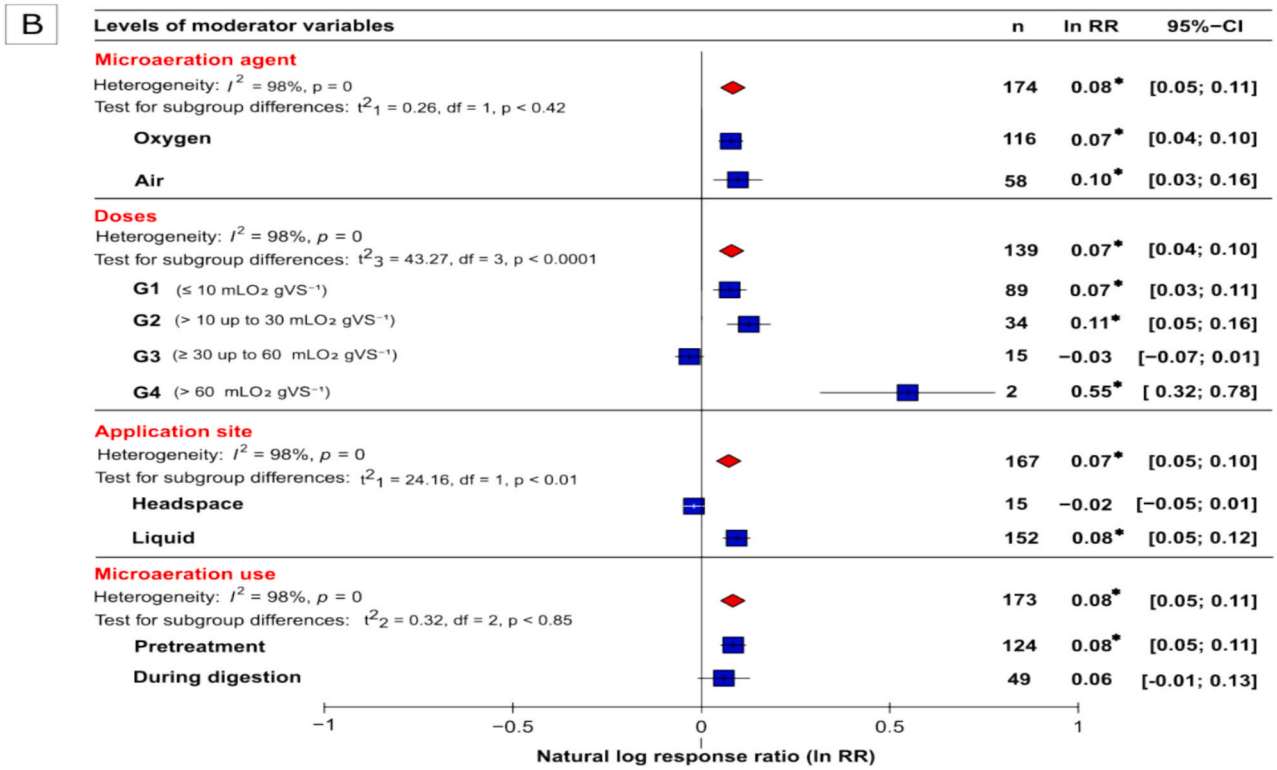


Fig. 5. The effects of microaeration on methane production in batch systems: (A) digestion parameters and (B) operational conditions. Abbreviations: n, number of observations; *Significant code ($p \leq 0.05$); 95 %-CI, 95 % confidence interval; t^2, test of heterogeneity among studies; I^2, reflects real differences in effect size and indicates its heterogeneity; df, degrees of freedom.

digestion systems. For biomass-based substrates, MA application resulted in a lnRR of 0.09 (95 % CI 0.05–0.12), corresponding to an average 9.0 % increase in methane production. This improvement can be attributed to the disruption of lignocellulosic structures (Fu et al., 2015), enhanced hydrolytic activity, and enrichment of bacteria capable of producing enzymes such as cellulase and xylanase (Zhu et al., 2022). For animal manure, the observed increase in methane production was even more pronounced (28.8 %), although the 95 % CI was relatively wide (0.12–0.39), reflecting the limited number of observations (Xu et al., 2024). Therefore, these conclusions should be interpreted with caution. Moreover, 62.5 % of the cases pertain specifically to chicken manure wastewater, a substrate already known to respond positively responses to various biological pretreatments (Anacleto et al., 2022). The beneficial effect of MA on this substrate may be linked to increased process stability and efficient H_2S removal.

These effects observed in animal manure–based systems can be explained by both metabolic and chemical mechanisms. From a metabolic perspective, MA can mitigate excessive VFA accumulation by stimulating facultative microorganisms and promoting the degradation of long-chain VFAs such as propionate and butyrate (Mahdy et al., 2020; Song et al., 2020). In particular, syntrophic acetogenic bacteria such as *Syntrophomonas* oxidise intermediate compounds in close association with hydrogenotrophic methanogens, thereby maintaining a low hydrogen partial pressure and facilitating the conversion of VFAs into acetate (Song et al., 2020). In parallel, *in situ* H_2S oxidation alters sulfur speciation and limits the precipitation of ferrous sulfide, thereby increasing the bioavailability of soluble iron. Given that iron is an essential enzymatic cofactor in methanogenic pathways, this process may enhance hydrogenotrophic methanogenesis, particularly in iron-dependent genera such as *Methanobacterium* (Song et al., 2020).

Co-digestion enhances methane production through synergistic interactions among substrates (Chuenchart et al., 2025) and represents an efficient strategy for simultaneously treating diverse organic materials within a single system. In this context, the application of MA resulted in an average 18.8 % increase in methane production (95 % CI 0.12–0.23) compared with conventional AD. Although based on a limited number of observations, these results indicate that MA may further improve methane yield under co-digestion conditions.

Most of the studies in this category involved agricultural residues, including lignocellulosic materials and animal manures, where MA improved substrate degradation and solubilisation. Under microaerobic conditions, the partial oxidation of complex polymers can stimulate the activity of hydrolytic and fermentative microorganisms, accelerating the depolymerisation of lignocellulosic structures and increasing the release of soluble intermediates, particularly VFAs (Zhan et al., 2023). In addition, MA has been associated with increased formation of humic-like substances, which can act as redox-active mediators facilitating extracellular electron transfer between syntrophic bacteria and methanogenic archaea (Jin et al., 2025). Collectively, these processes may enhance interspecies electron transfer and improve the efficiency of methanogenic pathways, thereby contributing to higher methane production during co-digestion.

In contrast, the application of MA to municipal solid waste produced a non-significant negative effect on methane production (ln RR = –0.02). More than 80 % of the studies in this group applied MA exclusively to the reactor headspace, which likely limited direct microbial contact and reduced its impact on hydrolytic processes (Song et al., 2021). The application of MA to sludge had a small non-significant positive effect on methane production (ln RR = 0.05), with a wide CI that included zero, indicating high inter-study variability and a limited sample size. In several of these studies, continuous air supply may have contributed to both biogas dilution and VFA accumulation during the mid-phase of digestion (Morello et al., 2021), underscoring the need to optimise oxygen input.

Temperature was also a key determinant of MA performance. Under mesophilic conditions, MA significantly increase methane production by 8.3 % (95 % CI 0.05–0.11), whereas there was no significant effect under thermophilic conditions (95 % CI 0.00–0.06). This reduced effectiveness at higher temperatures may be partly explained by the lower solubility of oxygen – which decreases exponentially between 273 and 400 K (Tromans, 1998) – and reduced stability of acetoclastic methanogens and a shift towards hydrogenotrophic methanogenesis (Fu et al., 2016; He et al., 2025). Given that MA has been associated with enhanced syntrophic interactions favouring hydrogenotrophic methanogens (Zhu et al., 2022), its relative benefit may diminish when this pathway predominates under thermophilic conditions.

Most of the batch-fed studies were conducted at the bench scale, with 65.8 % of the cases applying MA based on the solid content of the feed, underscoring the role of VS in shaping the effectiveness of MA. Only the VS2 subgroup (with a VS concentration of 10–30 g L^{-1}) showed a significant increase in methane production (17.9 %). This suggests that substrate concentration and associated mass transfer limitations strongly influence MA efficiency. As MA enhances the hydrolysis of complex organic matter, its effectiveness depends on the interaction between oxygen, microorganisms, and particulate substrates. Increased viscosity in systems with a high concentration of solids reduces diffusion and limits the transport of substrates and gases within the reactor (Abbassi-Guendouz et al., 2012). Consequently, oxygen supplied during MA may be distributed less effectively, given that effective MA depends on balanced oxygen transfer and microbial accessibility (Chuenchart et al., 2025). In addition, in some systems with high VS concentrations, oxygen was supplied at a relatively low level, which may have further restricted the observable effects (Fig. 4).

Across the assessed operational parameters (Fig. 5B), MA showed a consistent overall positive effect (lnRR = 0.08, 95 % CI 0.05–0.11), although there was substantial heterogeneity among the included studies. There was a slightly lower effect size for the oxygen input and application strategy with an identical lnRR (0.07) but slightly different 95 % CIs (0.04–0.10 and 0.05–0.10, respectively). Regarding the oxygen source, pure oxygen increased methane production by 7.0 %, while air led to a 10.2 % increase with greater variability (95 % CI 0.03–0.16), likely due to N_2-induced biogas dilution (Oliveira et al., 2021a; Silva et al., 2022).

Methane production differed significantly based on the oxygen input subgroups, with an increase of 6.9 % (G1), 11.1 % (G2), and 72.7 % (G4). However, the results for the G4 subgroup should be interpreted with caution, as it included only two studies and exhibited a wide 95 % CI (0.32–0.78). While insufficient oxygen supply may yield negligible effects, excessive oxygen can induce oxidative stress through the formation of reactive oxygen species (ROS), which can damage cellular components including membranes, proteins, and DNA (Nguyen and Khanal, 2018). In addition, carbon dioxide (CO_2) production may increase due to aerobic respiration by facultative microorganisms, inhibition of obligate anaerobes – particularly methanogens – and biogas dilution. In contrast, controlling the oxygen input may stimulate hydrolytic and facultative microorganisms, enhancing the degradation of complex organic matter and accelerating the initial stages of AD (Ullah et al., 2025).

The MA application strategy also significantly influenced methane production. Direct O_2 injection into the liquid phase resulted in a significant 8.8 % increase (95 % CI 0.05–0.12), whereas headspace application had a negative non-significant effect. Although headspace MA is effective for H_2S removal from biogas (> 90 %) (Díaz et al., 2010; Díaz et al., 2011; Krayzelova et al., 2015), its impact on digestion performance is limited by gas–liquid mass transfer constraints. Oxygen must first dissolve before interacting with microbial communities, reducing its bioavailability for hydrolytic and facultative microorganisms. This limitation becomes more pronounced in systems with higher organic loads, where gas–liquid transfer is further restricted (Nguyen and Khanal, 2018; Sheets et al., 2015).

Microaeration applied as a pretreatment resulted in an 8.5 % increase in methane production (95 % CI 0.05–0.11). This effect is likely

driven by partial oxidation and enzymatic breakdown of complex substrates prior to digestion, thereby increasing the availability of soluble intermediates for anaerobic digestion (Li et al., 2025). Moreover, pretreatment is less sensitive to variations in oxygen input, as oxygen is not directly introduced into the digester, thereby avoiding direct exposure of methanogens to oxygen and minimising the risks of inhibition and biogas dilution (Ullah et al., 2025). In contrast, application of MA during digestion requires stricter control of oxygen concentration and distribution to avoid process instability.

Overall, the findings indicate that MA can enhance methane production in batch systems; however, its effectiveness is strongly dependent on the substrate characteristics and operational conditions. The observed heterogeneity across the included studies and the limited number of observations in some subgroups reflects differences in oxygen input, application strategy, and reactor configuration. While the general trend is positive, optimisation of MA implementation remains essential to ensure consistent performance improvements in methane production.

3.2.2. The effect of MA on methane production in continuous feeding systems

The forest plots in Fig. 6 show the effects of MA on methane production under continuous feeding conditions, considering substrate characteristics (Fig. 6A) and operational parameters (Fig. 6B). The overall effect across substrate types was the only significant result (lnRR = 0.06, 95 % CI − 0.02 to 0.10), indicating a generally weaker response compared with batch systems and suggesting that the operational mode strongly influences MA effectiveness.

Among the substrate categories, only biomass exhibited a significant response, with a 6.6 % increase in methane production (95 % CI 0.02–0.11), despite the limited number of observations. This subgroup consisted exclusively of lignocellulosic substrates, for which MA may enhance the degradation of recalcitrant fractions. The reported mechanisms increased enzymatic activity (e.g., cellulases and xylanases), enhanced formation of humic-like substances, and improved hydrolysis of complex polymers (Zhu et al., 2022). These effects can contribute to improved process stability under high organic loading rates by accelerating VFA conversion, thereby increasing carbon recovery and volatile solids removal (Nguyen et al., 2019).

There was no overall significant effect for the VS concentration. However, for the VS1 subgroup, MA resulted in a modest but significant 5.0 % increase in methane production. This response differs from that observed in batch systems and may reflect operational characteristics specific to continuous systems. Notably, 97.5 % of the continuous studies were conducted at the demonstration, pilot, or large scale, where oxygen distribution may become more limited. Under these conditions, lower solid concentrations reduce viscosity and improve hydrodynamic conditions, facilitating oxygen diffusion and enhancing oxygen diffusion into the liquid phase (Abbassi-Guendouz et al., 2012; Chuenchart et al., 2025). Improved mass transfer under these conditions may therefore partly explain the observed benefit of MA in the VS1 subgroup.

Regarding operational parameters (Fig. 6B), the effects of the MA application site, agent type, and strategy were generally low and not significant, with consistently high heterogeneity across the included studies. Although there was no overall effect for the MA agent, there was a significant 12.8 % increase in methane production in the G4 subgroup (ln RR 0.12, $p \leq 0.05$). Similarly, oxygen introduction into the liquid phase resulting in a modest but significant 5.8 % increase in methane production, underscoring the importance of effective O_2 transfer in continuous reactors. In contrast, MA applied exclusively during digestion did not produce a significant effect, despite accounting for the largest number of observations.

When comparing batch and continuous systems, only a limited number of conditions – biomass, the G4 subgroup (with a limited number of experimental cases), and application of oxygen in the liquid phase – led to a significant positive response to MA across systems. While higher oxygen input appears to be associated with increased methane production, these findings should be interpreted with caution due to the small number of studies and potential variability in application strategies.

In batch systems, the promising effects of increased oxygen input were frequently associated with pretreatment strategies (Meng et al., 2015), where oxygen exposure is temporally separated from methanogenic activity. In contrast, continuous systems typically operate under steady-state feeding conditions, where oxygen is continuously supplied at a low intensity. Under these conditions, oxygen is rapidly consumed by facultative microorganisms involved in substrate degradation, thereby maintaining low dissolved oxygen concentrations and preventing inhibition of methanogens (Krayzelova et al., 2015; Oliveira et al., 2021b; Ullah et al., 2025).

The weaker and more variable response observed in continuous systems can therefore be attributed primarily to operational design. Many of the included studies were conducted at the pilot or large scale, where MA was often implemented primarily for H_2S removal rather than process intensification. In approximately half of the continuous systems, oxygen was applied to the headspace, which, while effective for desulfurisation, limits gas–liquid mass transfer and reduces its interaction with the anaerobic microbiome (Nguyen and Khanal, 2018; Sheets et al., 2015).

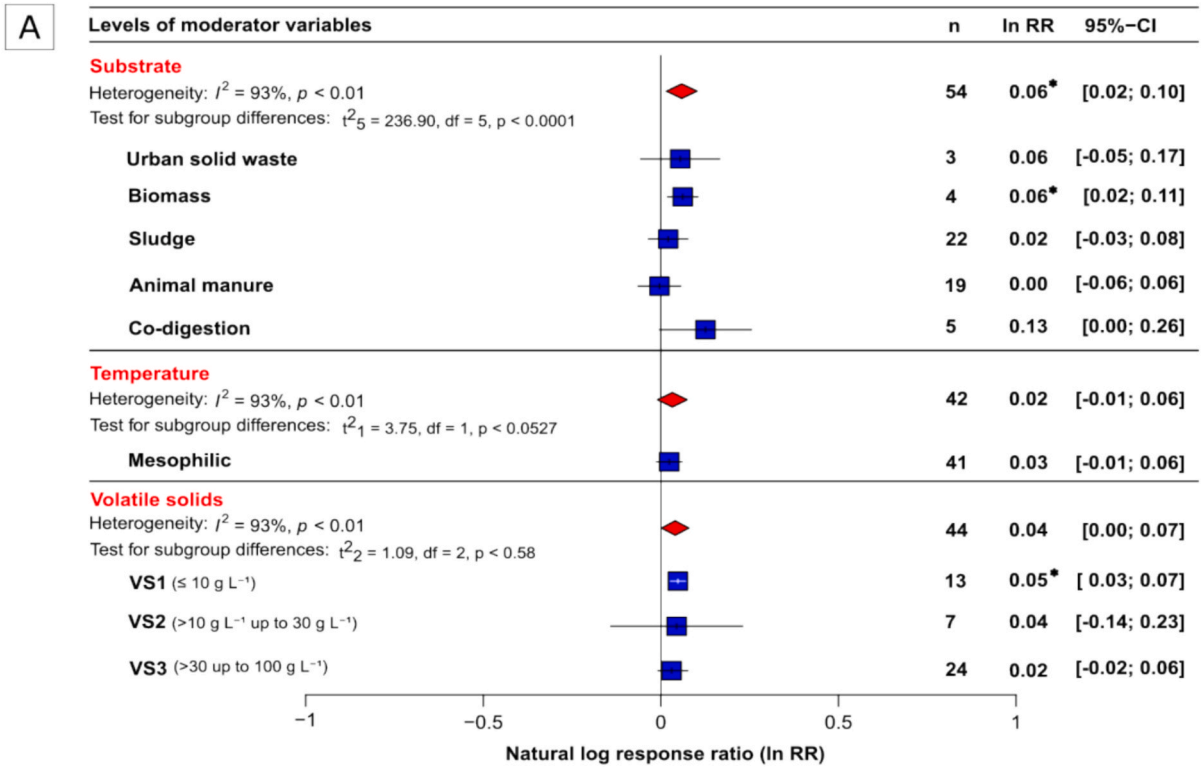
In addition, nearly 50 % of continuous studies operated under extended retention times (>100 days) and variable application strategies, often under stabilised conditions or imposed stress scenarios such as high organic loading rates. In such cases, MA has been reported to support hydrolytic activity and to improve process resilience (Zhu et al., 2022). However, differences in MA application strategies further complicate direct comparison between studies.

Specifically, more than 85 % of continuous experiments applied MA based on the feed volume, reactor volume, hydraulic retention time, or biogas production, in contrast to batch studies where oxygen input is more commonly normalised to the solid content. These inconsistencies in normalisation affect the comparability of oxygen supply intensity across studies. For example, Silva et al. (2022) demonstrated that maintaining a constant oxygen loading per feed volume under increasing organic loads can lead to biogas dilution effects, emphasising that both the oxygen input strategy and application intensity are critical determinants of process performance and operational stability.

3.3. Limitations, implications, and future perspectives

Despite the robustness of this systematic review and meta-analysis, some limitations should be acknowledged. First, the inclusion criteria established to ensure data consistency and comparability inevitably restricted the number of eligible studies on MA. In particular, the standardisation of oxygen input intensity required conversion to a common metric (mL O_2 g VS^{-1}). While this approach enabled cross-study comparison, it also resulted in the exclusion of studies that did not report sufficient information for conversion, especially those relying on alternative control metrics such as the ORP. In addition, incomplete reporting of key experimental details, including the number of replicates and measures of variability, further limited data inclusion in the meta-analysis.

MA is known to influence key stages of AD, particularly hydrolysis and syntrophic metabolism, by stimulating facultative and hydrolytic microorganisms and enhancing the degradation of complex substrates into soluble intermediates such as VFAs (Zhan et al., 2023; Zhu et al., 2022). Additional proposed mechanisms, including improved trace metal bioavailability via *in situ* H_2S oxidation and enhanced interspecies electron transfer, may further contribute to methanogenic activity (Di et al., 2025). However, the present synthesis indicates that the effectiveness of MA is strongly dependent on the interaction between substrate characteristics and operational conditions, with more consistent benefits observed in hydrolysis-limited systems such as lignocellulosic biomass, animal manure, and co-digestion processes.



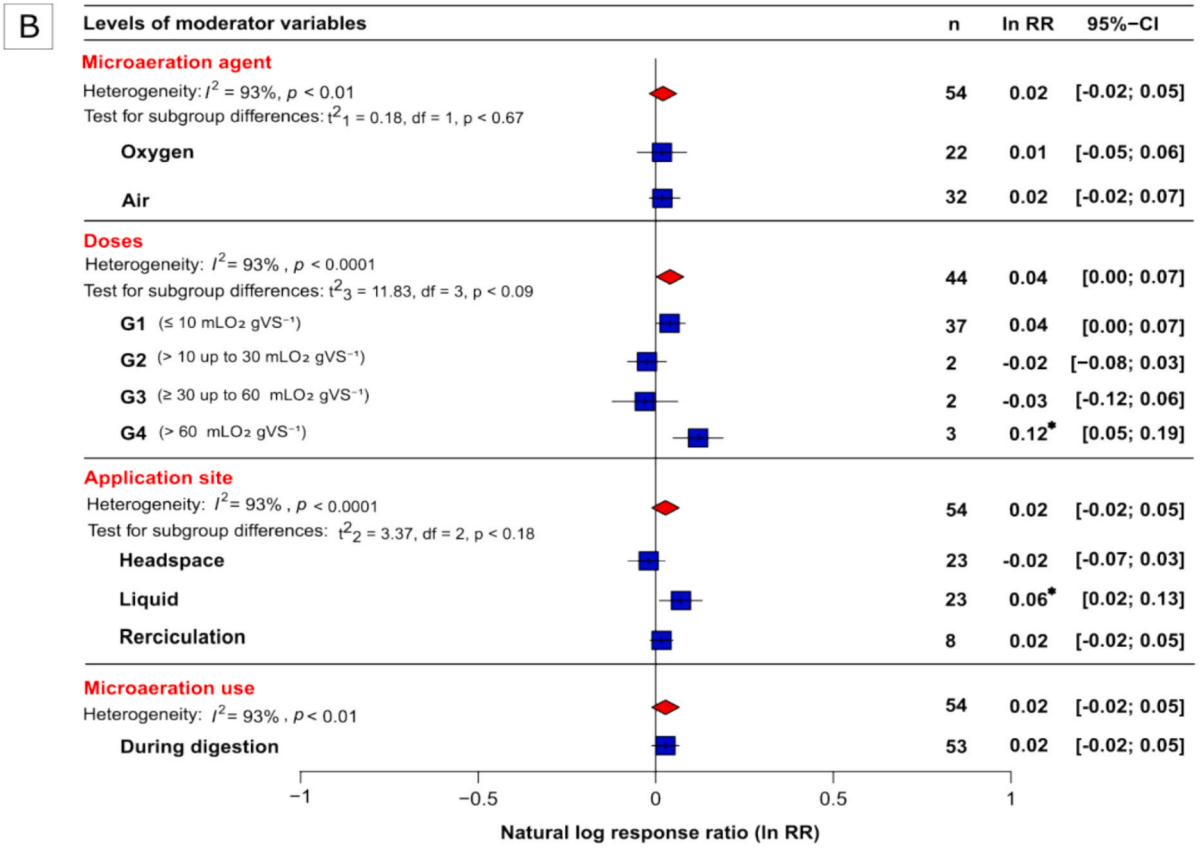


Fig. 6. The effect of microaeration on methane production in continuous systems: (A) digestion parameters (substrate type, temperature, and the volatile solid concentration) and (B) operational conditions. Abbreviations: n, number of observations; *Significant code ($p \leq 0.05$); 95 %-CI, 95 % confidence interval; t^2, test of heterogeneity among studies; I^2, reflects real differences in effect size and indicates its heterogeneity; df: degrees of freedom.

From an engineering perspective, these findings suggest that MA should not be considered a universally applicable strategy, but rather one that requires configuration-specific optimisation. Its application appears most effective under moderate solid concentrations and in systems where oxygen is delivered directly to the liquid phase, whereas performance is often reduced in systems with high solid concentration or when oxygen is applied via the headspace, due to mass transfer limitations. In continuous systems, MA may contribute more reliably to process stability than to enhancement of the methane yield, particularly under conditions of high organic loading or VFA accumulation. Overall, identifying the operational windows associated with positive responses provides a structured basis for reactor design and scale-up considerations.

Future research should focus on optimising the oxygen input strategies in continuous systems to improve scalability from laboratory to full-scale applications. Greater attention should also be given to routine monitoring of accessible process indicators, such as alkalinity, to better capture system stability (Chuenchart et al., 2025). In parallel, mechanistic studies integrating microbial community dynamics, electron transfer pathways, and enzymatic activity under varying MA application strategies are still needed to refine process understanding. Standardisation of how oxygen input is reported – through unified metrics or complete operational descriptors (e.g., the ORP, the solid content, as well as reactor and feed volumes) – would substantially improve comparability across studies and strengthen future meta-analyses. Finally, more consistent reporting of experimental design elements, including replication and measures of variability, is essential to enhance the reliability and interpretability of future evidence syntheses.

4. Conclusions

MA has gained increasing attention since 2010. Based on the findings, it has been predominantly applied as a pretreatment (52.8 %) or during digestion (46.5 %), mainly at the bench scale (69.4 %), and with oxygen delivered directly to the liquid phase. Among the evaluated substrates, biomass was the most frequently studied (40.2 %) and, together with animal manure, exhibited the strongest responses to MA application. In batch systems, biomass and animal manure increased methane production by 9.0 % and 28.8 %, respectively, particularly under mesophilic conditions and with intermediate VS concentrations (10–30 g L^{-1}, +17.9 %).

The effectiveness of MA is strongly dependent on operational parameters, including oxygen input, application site, and mass transfer efficiency, with direct liquid injection increasing methane production by 5.8 %. In continuous systems, the overall effects were lower and often constrained by operational configurations such as headspace injection; nevertheless, biomass still showed a modest but significant improvement in methane production (6.6 %).

Overall, MA demonstrates clear potential to enhance the key rate-limiting steps of AD. However, the high heterogeneity across studies and the limited number of large-scale investigations highlight the need for optimised and standardised operational protocols. Future development of MA for the organic waste and wastewater treatment sector will require targeted pilot and large-scale studies, improved normalisation of oxygen input relative to biodegradable fractions, and integration of monitoring and control strategies to enable its robust and scalable application.

CRediT authorship contribution statement

Laís Veloso Febroni: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **André Pereira Rosa:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alisson Carraro Borges:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.135162.

Data availability

https://data.mendeley.com/datasets/fwwszfp78n/1

Dataset supporting the article "Microaeration enhances methane production in anaerobic digestion: A systematic review and meta-analysis" (Reference data) (Mendeley Data)

References

Anacleto, T.M., Oliveira, H.R., Diniz, V.L., de Oliveira, V.P., Abreu, F., Enrich-Prast, A., 2022. Boosting manure biogas production with the application of pretreatments: a meta-analysis. J. Clean. Prod. 362 (9), 132292. https://doi.org/10.1016/j.jclepro.2022.132292.

Abbassi-Guendouz, A., Brockmann, D., Trably, E., Dumas, C., Delgenès, J.P., Steyer, J.P., Escudié, R., 2012. Total solids content drives high solid anaerobic digestion via mass transfer limitation. Bioresour. Technol. 111, 55–61. https://doi.org/10.1016/j.biortech.2012.01.174.

Bao, H., Yang, H., Zhang, H., Liu, Y., Shen, M., 2020. Improving methane productivity of waste activated sludge by ultrasound and alkali pretreatment in microbial electrolysis cell and anaerobic digestion coupled system. Environ. Res. 180, 108863. https://doi.org/10.1016/j.envres.2019.108863.

Bernardes, R.C., Botina, L.L., Araújo, R.S., Guedes, R.N.C., Martins, G.F., Lima, M.A.P., 2022. Artificial intelligence-aided meta-analysis of toxicological assessment of agrochemicals in bees. Front. Ecol. Evol. 10, 845608. https://doi.org/10.3389/fevo.2022.845608.

Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2020. Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388. https://doi.org/10.1016/j.scitotenv.2019.136388.

Chetawan, W., Saritpongteeraka, K., Palamanit, A., Chaiprapat, S., 2021. Practical approaches for retrofitting plug flow digester and process control to maximize hydrolysis and methane yield from piggery waste. J. Environ. Chem. Eng. 9, 105620. https://doi.org/10.1016/j.jece.2021.105620.

Chuenchart, W., Sawaya, C., Surendra, K.C., Smith, A.L., Khanal, S.K., 2025. In-situ intermittent micro-aeration in food waste and sewage sludge anaerobic co-digestion: performance, stability, and microbial dynamics. Bioresour. Technol. 427, 132398. https://doi.org/10.1016/j.biortech.2025.132398.

DerSimonian, R., Laird, N., 1986. Meta-analysis in clinical trials. Control. Clin. Trials 7, 177–188.

Di, Y., Hou, W., Bian, C., Zheng, T., Xiao, B., Li, L., 2025. Enhancing anaerobic digestion of swine manure using microbial electrolysis cell and microaeration. Chem. Eng. J. 514, 163319. https://doi.org/10.1016/j.cej.2025.163319.

Diaz, I., Lopes, A.C., Perez, S.I., Fdz-Polanco, M., 2010. Performance evaluation of oxygen, air and nitrate for the microaerobic removal of hydrogen sulphide in biogas from sludge digestion. Bioresour. Technol. 101, 7724–7730. https://doi.org/10.1016/j.biortech.2010.04.062.

Díaz, I., Pérez, S.I., Ferrero, E.M., Fdz-Polanco, M., 2011. Effect of oxygen dosing point and mixing on the microaerobic removal of hydrogen sulphide in sludge digesters. Bioresour. Technol. 102, 3768–3775. https://doi.org/10.1016/j.biortech.2010.12.016.

Duan, Y., Wang, Z., Ganeshan, P., Sar, T., Xu, S., Rajendran, K., Sindhu, R., Binod, P., Pandey, A., Zhang, Z., Taherzadeh, M.J., Awasthi, M.K., 2025. Anaerobic digestion in global bio-energy production for sustainable bioeconomy: potential and research

challenges. Renew. Sustain. Energy Rev. 208, 114985. https://doi.org/10.1016/j.rser.2024.114985.
Feng, S., Ngo, H.H., Guo, W., Chang, S.W., Nguyen, D.D., Liu, Y., Zhang, S., Vo, H.N.P., Bui, X.T., Hoang, B.N., 2022. Volatile fatty acids production from waste streams by anaerobic digestion: a critical review of the roles and application of enzymes. Bioresour. Technol. 359, 127420. https://doi.org/10.1016/j.biortech.2022.127420.
Fu, S.-F., Shi, X.-S., Wang, F., Guo, R.-B., 2016. Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure. Chem. Eng. J. 287, 523–528. https://doi.org/10.1016/j.cej.2015.11.070.
Fu, S., Shi, X., Wang, F., Yuan, X., Guo, R.-B., 2015. Comparison of thermophilic microaerobic and alkali pretreatment of sugarcane bagasse for anaerobic digestion. RSC Adv. 5, 63903–63908. https://doi.org/10.1039/c5ra10717b.
Girotto, F., Peng, W., Rafieenia, R., Cossu, R., 2018. Effect of aeration applied during different phases of anaerobic digestion. Waste Biomass Valoriz. 9, 161–174. https://doi.org/10.1007/s12649-016-9785-9.
He, H., Chen, X., Peng, Z., Zhang, Z., Ren, X., Ma, S., Khan, M.U., Cui, Z., Yuan, X., 2025. A review of temperature and key parameters influencing the hydrolysis–methanogenesis balance in anaerobic digestion. Fuel 394, 134927. https://doi.org/10.1016/j.fuel.2025.134927.
Hedges, L.V., Gurevitch, J., Curtis, P.S., 1999. The meta-analysis of response ratios in experimental ecology. Ecology 80, 1150–1156. https://doi.org/10.2307/177062.
Huilinir, C., Pages-Diaz, J., Vargas, G., Vega, S., Lauzurique, Y., Palominos, N., 2023. Microaerobic condition as pretreatment for improving anaerobic digestion: a review. Bioresour. Technol. 384, 129249. https://doi.org/10.1016/j.biortech.2023.129249.
Jackson, D., White, I.R., Riley, R.D., 2012. Quantifying the impact of between-study heterogeneity in multivariate meta-analyses. Stat. Med. 31, 3805–3820. https://doi.org/10.1002/sim.5453.
Jin, Z., Yu, Q., Zhao, Z., Zhang, Y., 2025. Insight into the role of intermitente microaeration in enhancing lignocellulose-containing anaerobic digestion and humic substances formation. Chem. Eng. J. https://doi.org/10.1016/j.cej.2025.166500.
Kraakman, N.J.R., Diaz, I., Fdz-Polanco, M., Munoz, R., 2023. Large-scale micro-aerobic digestion studies at municipal water resource recovery facilities for process-integrated biogas desulfurization. J. Water Process Eng. 53, 103643. https://doi.org/10.1016/j.jwpe.2023.103643.
Krayzelova, L., Bartacek, J., Díaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Biotechnol. 14, 703–725. https://doi.org/10.1007/s11157-015-9386-2.
Li, X., Yan, Y.-J., Wu, H.-M., Gadow, S.I., Jiang, H., Kong, Z., Hu, Y., 2025. Enhancing mesophilic methanogenesis in oleate-rich environments through optimized micro-aeration pretreatment. Waste Manag. 193, 171–179. https://doi.org/10.1016/j.wasman.2024.12.005.
Lim, J.W., Wang, J.-Y., 2013. Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste. Waste Manag. 33, 813–819. https://doi.org/10.1016/j.wasman.2012.11.013.
Mahdy, A., Song, Y., Salama, A., Qiao, W., Dong, R., 2020. Simultaneous H_2S mitigation and methanization enhancement of chicken manure through the introduction of the micro-aeration approach. Chemosphere 253, 126687. https://doi.org/10.1016/j.chemosphere.2020.126687.
Meng, Y., Jost, C., Mumme, J., Wang, K., Linke, B., 2015. Oxygen tolerance capacity of upflow anaerobic solid-state (UASS) with anaerobic filter (AF) system. J. Environ. Sci. (China) 45, 200–206. https://doi.org/10.1016/j.jes.2016.01.009.
Montalvo, S., Huilinir, C., Ojeda, F., Castillo, A., Lillo, L., Guerrero, L., 2016. Microaerobic pretreatment of sewage sludge: effect of air flow rate, pretreatment time and temperature on the aerobic process and methane generation. Int. Biodeterior. Biodegrad. 110, 1–7. https://doi.org/10.1016/j.ibiod.2016.01.010.
Morello, R., Di Capua, F., Pontoni, L., Papirio, S., Spasiano, D., Fratino, U., Pirozzi, F., Esposito, G., 2021. Microaerobic digestion of low-biodegradable sewage sludge: effect of air dosing in batch reactors. Sustainability 13, 9869. https://doi.org/10.3390/su13179869.
Nguyen, D., Khanal, S.K., 2018. A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. Biotechnol. Adv. 36, 1971–1983. https://doi.org/10.1016/j.biotechadv.2018.08.007.
Nguyen, D., Wu, Z., Shrestha, S., Lee, P.-H., Raskin, L., Khanal, S.K., 2019. Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res. 166. https://doi.org/10.1016/j.watres.2019.115080.
Oliveira, M., Marques Mourão, J.M., Souza Silva, F.S., dos Santos, A.B., Pereira, E.L., 2021a. Effect of microaerophilic treatment on swine wastewater (SWW) treatment: engineering and microbiological aspects. J. Environ. Manag. 299, 113598. https://doi.org/10.1016/j.jenvman.2021.113598.
Oliveira, M., Marques Mourão, J.M., Marques de Oliveira, A.K., dos Santos, A.B., Pereira, E.L., 2021b. Microaerophilic treatment enhanced organic matter removal and methane production rates during swine wastewater treatment: a long-term engineering evaluation. Renew. Energy 180, 691–699. https://doi.org/10.1016/j.renene.2021.08.120.
Page, M.J., Moher, D., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McKenzie, J.E., 2021. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ 372, n160. https://doi.org/10.1136/bmj.n160.
R Core Team., 2024. *R: A language and environment for statistical computing* (Version 4.3.3) [Computer software]. R Foundation for Statistical Computing. https://www.R-project.org.
Rashvanlou, R.B., Rezaee, A., Farzadkia, M., Gholami, M., Kermani, M., 2020. Effect of micro-aerobic process on improvement of anaerobic digestion sewage sludge treatment: flow cytometry and ATP assessment. RSC Adv. 10, 35718–35728. https://doi.org/10.1039/d0ra05540a.
Rosas-Mendoza, E.S., Alvarado-Vallejo, A., Vallejo-Cantú, N.A., Snell-Castro, R., Martínez-Hernández, S., Alvarado-Lassman, A., 2021. Batch and semi-continuous anaerobic digestion of industrial solid citrus waste for the production of bioenergy. Processes 9, 648. https://doi.org/10.3390/pr9040648.
Rohatgi, A., 2023. WebPlotDigitizer, version 4.8. Available at: https://apps.automeris.io/wpd4/.
Sheets, J.P., Ge, X., Li, Y., 2015. Effect of limited air exposure and comparative performance between thermophilic and mesophilic solid-state anaerobic digestion of switchgrass. Bioresour. Technol. 180, 296–303. https://doi.org/10.1016/j.biortech.2015.01.011.
Silva, T.P., de Oliveira, M.G., Mourão, J.M.M., Possetti, G.R.C., Pereira, E.L., dos Santos, A.B., 2022. Bioenergy recovery potential from upflow microaerobic sludge blanket reactor fed with swine wastewater. Biochem. Eng. J. 187, 108675. https://doi.org/10.1016/j.bej.2022.108675.
Song, C., Li, W., Cai, F., Liu, G., Chen, C., 2021. Anaerobic and microaerobic pretreatment for improving methane production from paper waste in anaerobic digestion. Front. Microbiol. 12, 688290. https://doi.org/10.3389/fmicb.2021.688290.
Song, Y., Mahdy, A., Hou, Z., Lin, M., Stinner, W., Qiao, W., Dong, R., 2020. Air supplement as a stimulation approach for the in situ desulfurization and methanization enhancement of anaerobic digestion of chicken manure. Energy Fuels 34, 12606–12615. https://doi.org/10.1021/acs.energyfuels.0c01724.
Tromans, D., 1998. Temperature and pressure dependent solubility of oxygen in water: a thermodynamic analysis. Hydrometall. 48, 327–342. https://doi.org/10.1016/S0304-386X(98)00007-3.
Ullah, U., Alkorbid, A.S., Jalalah, M., Harraz, F.A., Alalawy, A.I., Hassan, S.H.A., Salama, E.S., 2025. Lab to large-scale applications of microaeration in anaerobic digestion: Biowaste digestibility, microbiome nexus, and reactor stability. J. Environ. Chem. Eng. 13, 117766. https://doi.org/10.1016/j.jece.2025.117766.
United Nations., 2015. Transforming our world: The 2030 agenda for sustainable development. United Nations. Available at: https://www.un.org/sustainabledevelopment/.
Van de Schoot, R., de Bruin, J., Schram, R., Zahedi, P., de Boer, J., Weijdema, F., et al., 2021. ASReview: active learning for systematic reviews [Computer software]. Utrecht University. https://asreview.ai.
Xu, J., Wang, L., Song, C., Jin, Y., Chen, C., Liu, G., 2024. A meta-analysis for effects of pretreatments on corn straw for methane production. Ind. Crop. Prod. 219, 118895. https://doi.org/10.1016/j.indcrop.2024.118895.
Yu, Q., Liu, R., Li, K., Ma, R., 2019. A review of crop straw pretreatment methods for biogas production by anaerobic digestion in China. Renew. Sustain. Energy Rev. 107, 51–58. https://doi.org/10.1016/j.rser.2019.02.020.
Zhan, Y., Zhu, J., Xiao, Y., Schrader, L.C., Wu, S.X., Robinson, N.A., Wang, Z., 2023. Employing micro-aeration in anaerobic digestion of poultry litter and wheat straw: batch kinetics and continuous performance. Bioresour. Technol. 368, 128351. https://doi.org/10.1016/j.biortech.2022.128351.
Zhu, R., Zhang, Y., Zou, H., Guo, R.-B., Fu, S.-F., 2022. The effects of micro-aeration on semi-continuous anaerobic digestion of corn straw with increasing organic loading rates. Renew. Energy 195, 1194–1201. https://doi.org/10.1016/j.renene.2022.06.106.